

Vitória Barin Pacela

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Education

Ph.D., Computer Science – Université de Montréal, Mila 2021 – present

Supervisor: Prof. [Simon Lacoste-Julien](#). Causal representation learning, disentanglement, interpretability.

M.Sc., Data Science – University of Helsinki 2019 – 2021

Thesis: Independent Component Analysis for Binary Data. Supervisor: Prof. [Aapo Hyvärinen](#)

B.Sc., Computer Science – University of Helsinki 2015 – 2019

Minors in Theoretical Physics and Mathematics/Statistics.

Experience

Cold Spring Harbor Laboratory Jun. 2025 – Sep. 2025

NeuroAI Summer Intern, New York, USA

Advised by Prof. [David Klindt](#). Investigated out-of-distribution generalization in the overcomplete setting via sparse coding as an alternative to sparse autoencoders; mentored an undergraduate researcher. Manuscript in preparation.

Meta – Fundamental AI Research (FAIR) Oct. 2022 – Oct. 2024

Visiting Researcher, Montréal, Canada (part-time)

Worked with Prof. [Pascal Vincent](#) on identifiability of quantized latent factors (CLeaR 2024) and a practical criterion for quantized disentanglement.

Earlier research experience: Mila (2019, streamflow forecasting under Prof. Yoshua Bengio); CERN Openlab (2018) and Caltech/CMS (2017), fast calorimeter simulation with GANs; Caltech (2016), photonic crystal characterization.

Selected Publications

- [V. Barin-Pacela*](#), [S. Joshi*](#), [I. Camacho](#), [S. Lacoste-Julien](#), [D. Klindt](#). *Stop Probing, Start Coding: Why Linear Probes and Sparse Autoencoders Fail at Compositional Generalisation*. Conference on Uncertainty in Artificial Intelligence (UAI) 2026.
- [V. Barin-Pacela](#), [K. Ahuja](#), [S. Lacoste-Julien](#), [P. Vincent](#). *Operationalizing Quantized Disentanglement*. 2025.
- [V. Barin-Pacela](#), [K. Ahuja](#), [S. Lacoste-Julien](#), [P. Vincent](#). *On the Identifiability of Quantized Factors*. Conference on Causal Learning and Reasoning (CLeaR) 2024.
- [A. Hyttinen](#), [V. Barin-Pacela](#), [A. Hyvärinen](#). *Binary Independent Component Analysis: A Non-stationarity-based Approach*. Conference on Uncertainty in Artificial Intelligence (UAI) 2022.
- [D. Belayneh](#), [F. Carminati](#), [A. Farbin](#), [B. Hooberman](#), [G. Khattak](#), [M. Liu](#), [J. Liu](#), [D. Olivito](#), [V. Barin Pacela](#), [M. Pierini](#), [A. Schwing](#), [M. Spiropulu](#), [S. Vallecorsa](#), [J-R. Vlimant](#), [W. Wei](#), and [M. Zhang](#). *Calorimetry with Deep Learning: Particle Identification and Simulation for Collider Physics*. *European Physical Journal C*, 80(7), 2020.

Awards

Mila EDI Scholarship, Women in AI (2024–2027, \$8,000/yr); Amii Upper Bound Talent Bursary (2025); Professor Cho Diversity Award, Mila (2021); Travel awards – ICML 2023, NeurIPS 2019, RIIAA LATAM 2024; Instituto TIM Selected Scholar (2015–2019, R\$57,600.)

Invited Talks

Bayes Plurinacional (2025); EIA University (2025); Bellairs Workshop on Causality (2025); Cold Spring Harbor Laboratory (2024); ISTA (2024); [Mila GFlowNet Workshop](#) (2023), [Instituto de Matemática Pura e Aplicada](#) (2023), FGV EMaP (2023).

Teaching

Representation Learning ([IFT6135](#), Mila/Université de Montréal) lectured by Professor [Aaron Courville](#). 2025.

Leadership

Supervised one undergraduate student and one early-stage PhD student.

Meta Women in AI Steering Committee (2023/2024).

Selected Service

Reviewer – AISTATS 2024/2025, CLeaR 2025, UAI 2025. Teaching Skills Committee, University of Helsinki (2020). CERN International Masterclasses, invited speaker (2017–2026). Created the Mila Library (2022–2023).

Miscellanea

Scientific Olympiads (2009–2014): 21 prizes in Brazilian nationals.

Languages: English, Portuguese, French.

Hatha & Vinyasa Yoga Teacher Training (380h) w/ [Sylvie Tremblay](#) at Yoga Sangha, 2026.